Design and Realization of an Affordable Continuous Sub-Cutaneous Insulin Infusion Pump for Precise Insulin Delivery

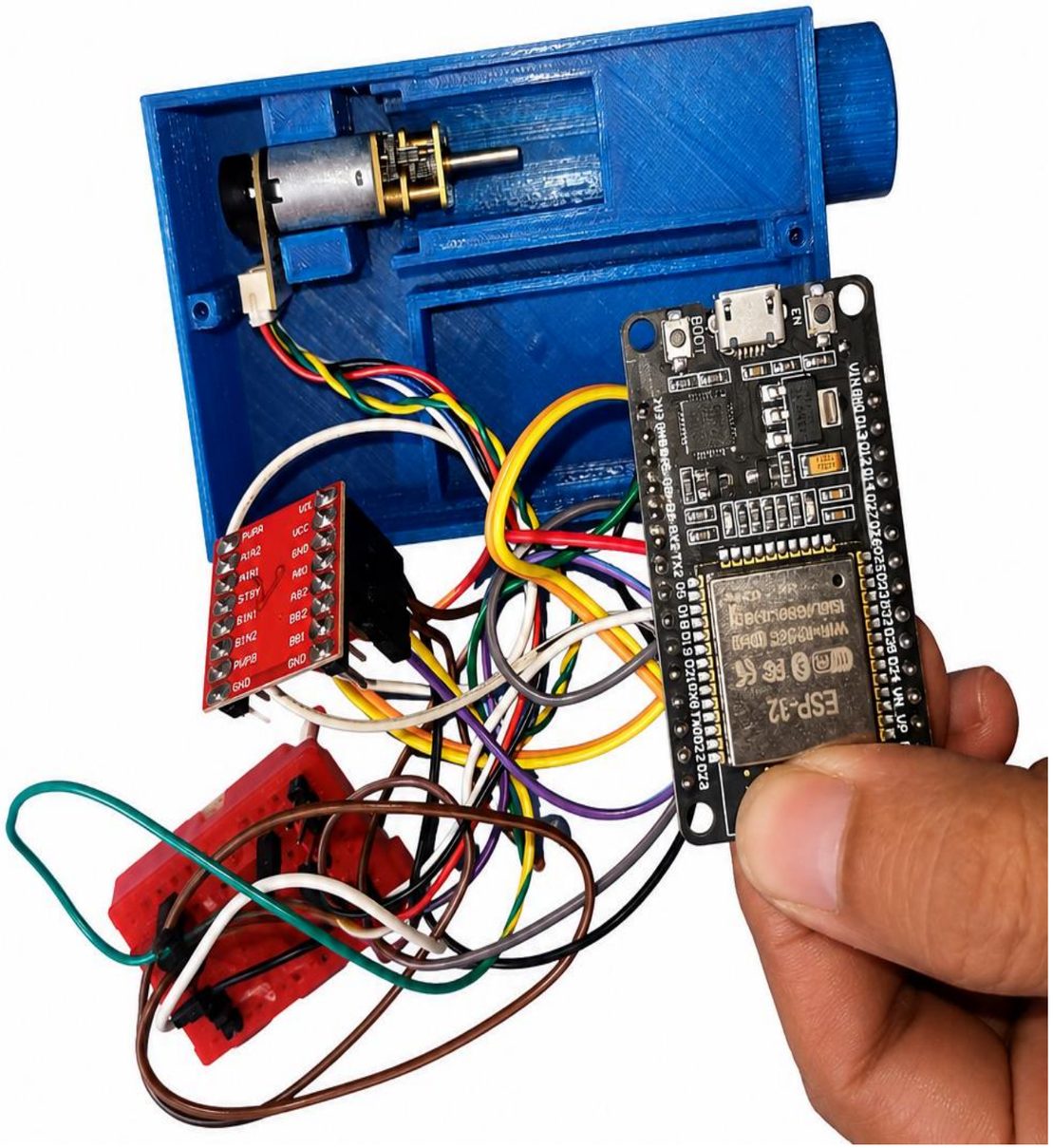
Team Members:

- Debjyoti Saha
- Eppa Mani preetham
- Harshaditya Punera

Project Mentor: Dr. Anirudh Nath



Introduction



The Clinical Challenge

Diabetes: A Global Crisis

- Type 1 Diabetes: Autoimmune destruction of pancreatic beta cells requiring lifelong insulin replacement
- Type 2 Diabetes: Progressive insulin resistance and declining beta-cell function
- 537 million adults worldwide affected, projected to reach 783 million by 2045

Risks of Poor Glycaemic Control

- **Hypoglycaemia:** Blood glucose <3.9 mmol/L causing confusion, seizures, unconsciousness
- **Hyperglycaemia:** Chronic elevation leading to retinopathy, nephropathy, neuropathy, cardiovascular disease
- **Precision imperative:** Tight glycaemic control (HbA1c <7%) reduces complications by 40-75%

Evolution of Insulin Delivery Systems

1920s-1980s

Syringes & Vials: Manual injection, poor precision, patient non-compliance, painful administration

1

1985-2000

Insulin Pens: Improved convenience and discrete dosing, but limited basal-bolus flexibility

2

2000-Present

Insulin Pumps: Continuous subcutaneous infusion, programmable basal rates, precise bolus delivery

3

Future

Closed-Loop Systems: Integration with continuous glucose monitoring for automated insulin delivery

4

Limitations of Manual Injection

Dawn Phenomenon: Early morning hyperglycaemia requires precise basal adjustment impossible with long-acting insulin

Compliance Burden: Multiple daily injections (4-6/day) reduce adherence, particularly in paediatric and geriatric populations

Dosing Inaccuracy: Pen increments of 0.5-1 IU insufficient for precise micro-dosing in children and insulin-sensitive patients

Insulin Dosing Principles

Insulin Types

- **Rapid-acting:** Onset 10-15 min, peak 1-2 hrs (lispro, aspart)
- **Short-acting:** Onset 30 min, peak 2-4 hrs (regular insulin)
- **Long-acting:** Duration 18-24+ hrs (glargine, degludec)

Concentration Standards *(*IU=Insulin Units) (1 IU = 0.0347 mg)*

- **U-100:** 100 IU/mL (standard)
- **U-200:** 200 IU/mL (concentrated formulations)
- **Critical distinction:** Volume ≠ Units. Precision requires µL-level control

Basal Insulin Delivery

Purpose: Physiologic mimicry of pancreatic background insulin secretion (0.5-1.0 IU/hr typical range)

Requirements: Programmable hourly rates to address circadian variation. Precision: ±0.025 IU/hr minimum

Bolus Insulin Delivery

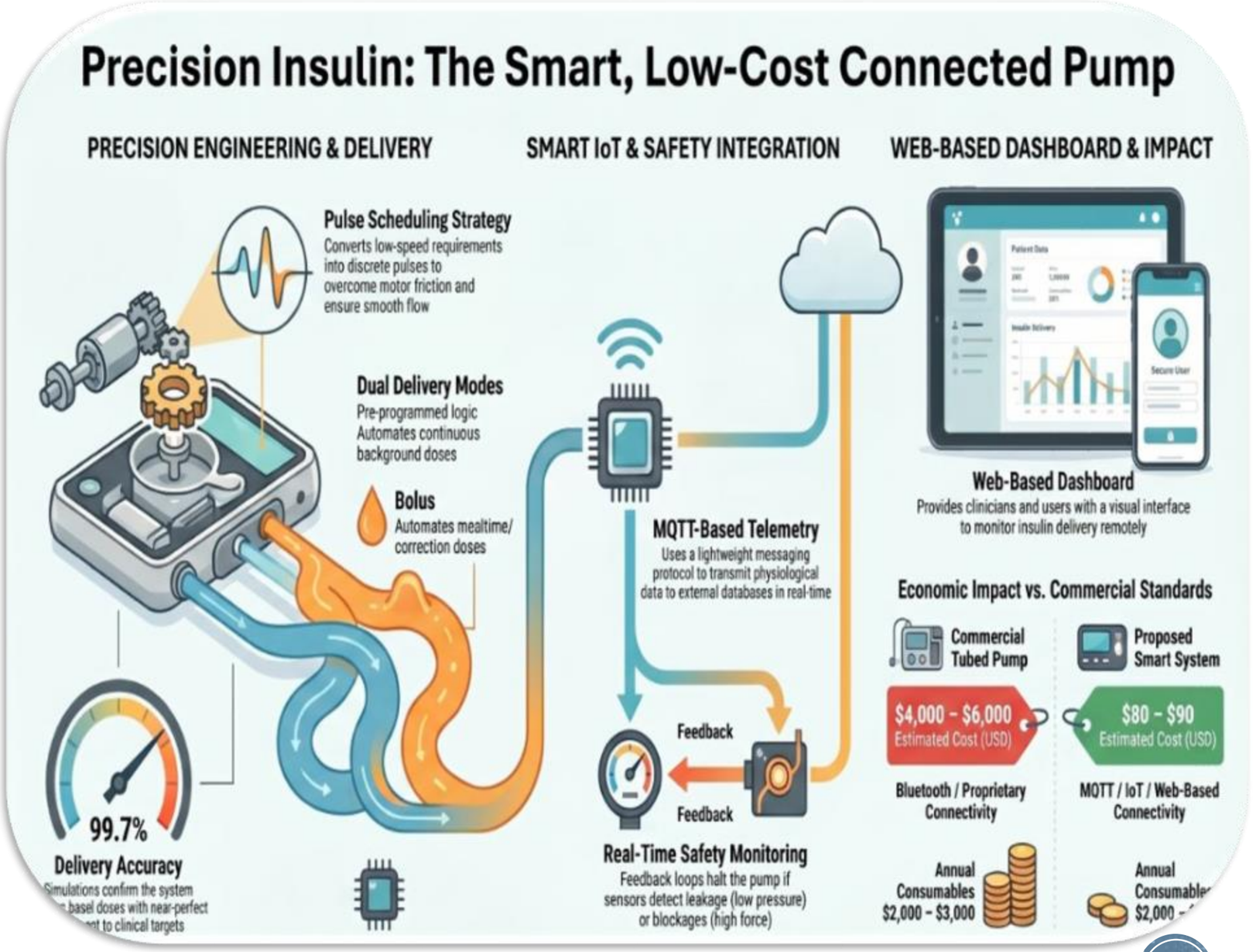
Purpose: Meal coverage and hyperglycaemia correction

Components: Carbohydrate ratio (1 IU per 10-15g CHO) plus correction factor (1 IU reduces glucose by 2-3 mmol/L)

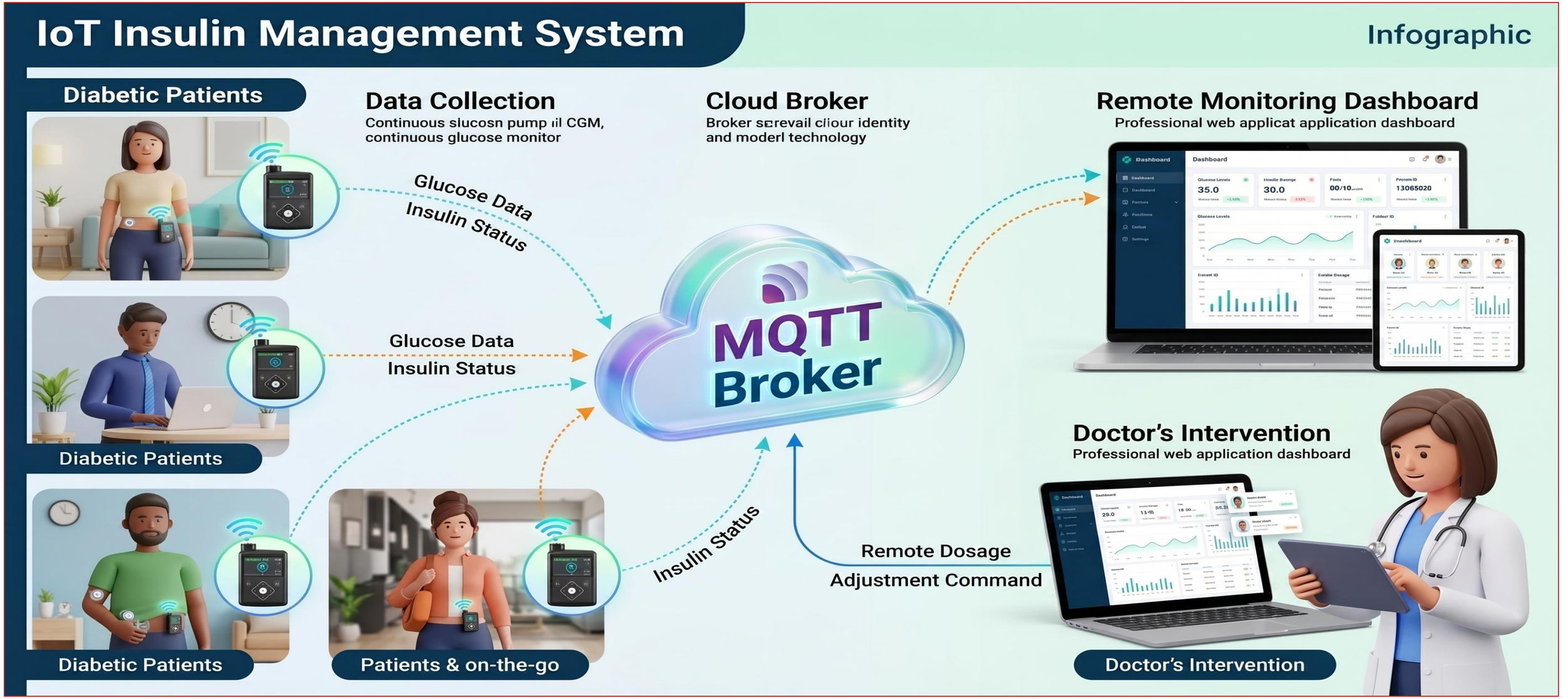
Requirements: Rapid delivery (0.05-15 IU), increment precision of 0.025 IU

Our Approach

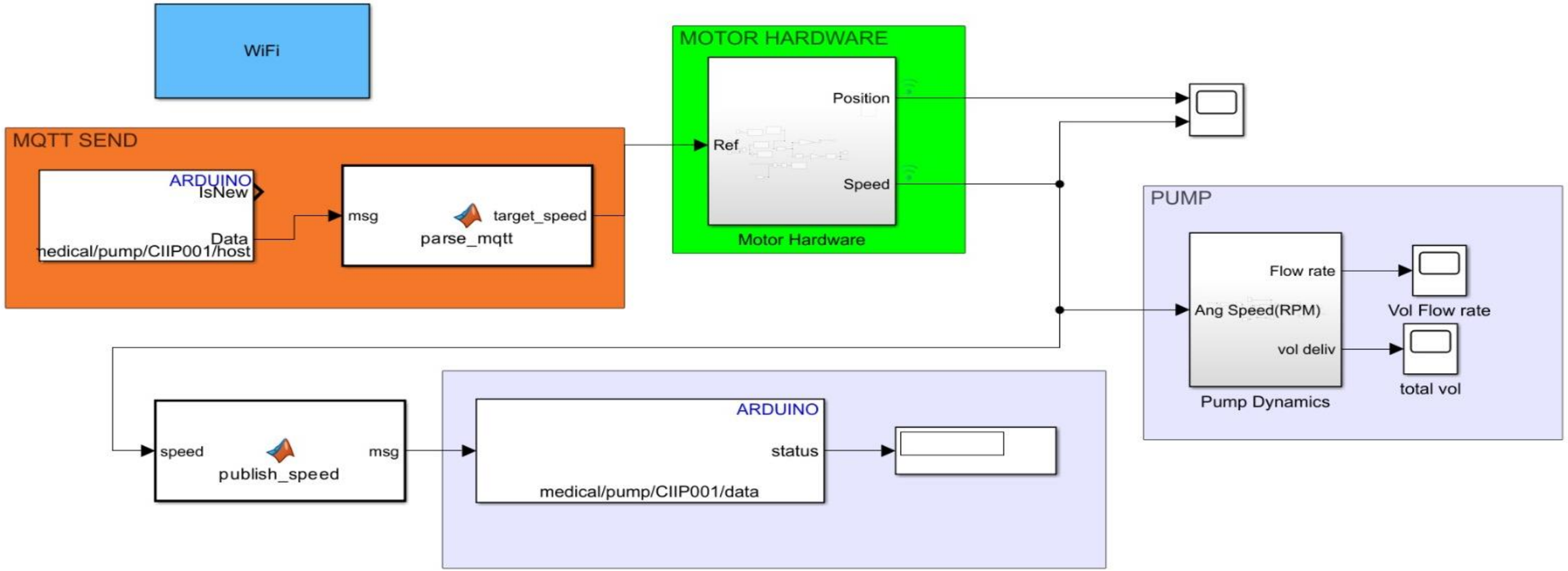
- Our approach is centered on developing a safe and high-precision insulin infusion pump using, with the simulation and results closely matching for use of a patient.
- We began by modelling the system on **MATLAB/Simulink** in last semester.
- In this semester our **1st prototype** is constructed, followed by the design of a closed-loop **PID controller** for precise speed regulation.
- To ensure patient safety, multiple protection layers are integrated into the model.
- A **WebApp** is designed to control and view real time data.
- Finally, the integrated model is validated for basal and bolus delivery.



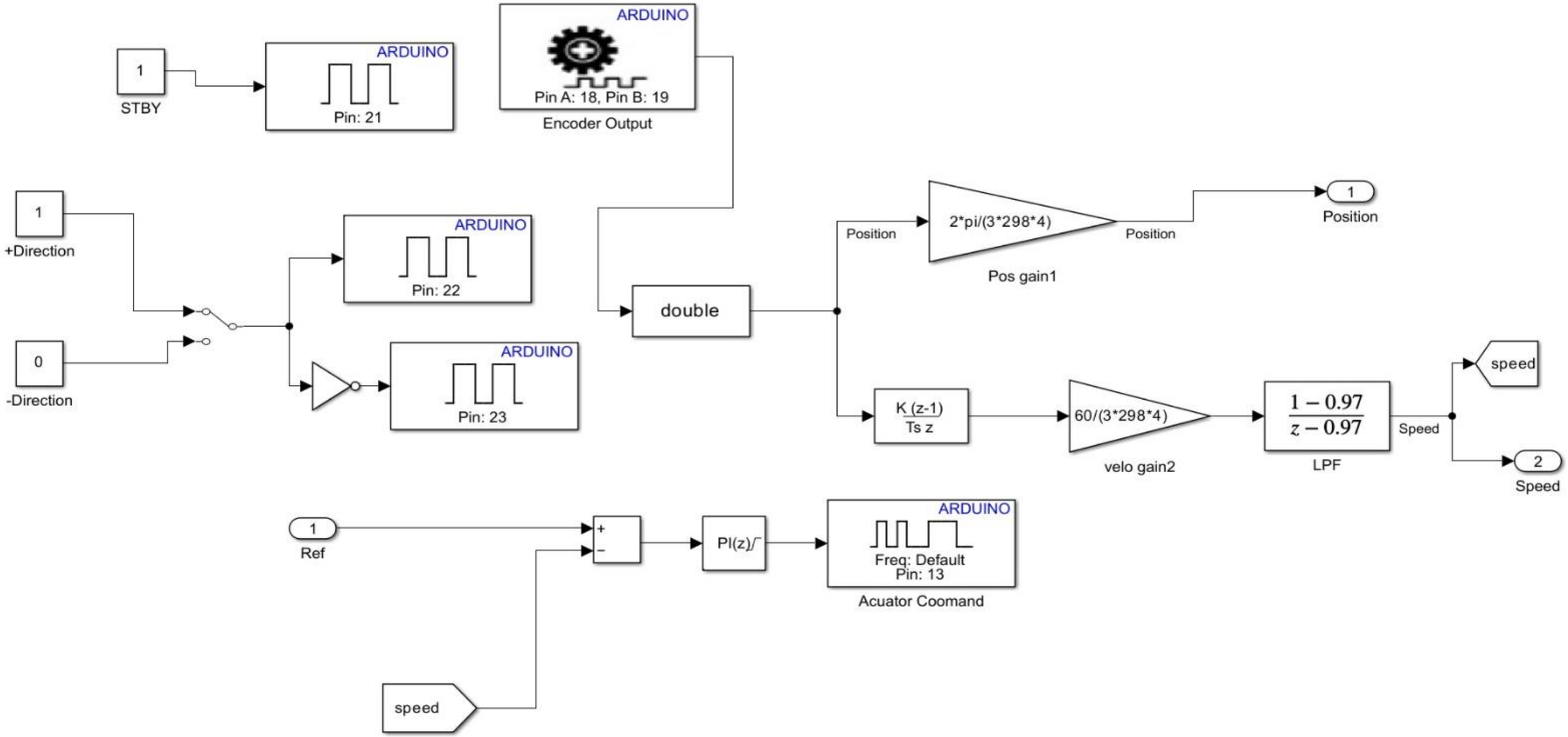
Overall Architecture



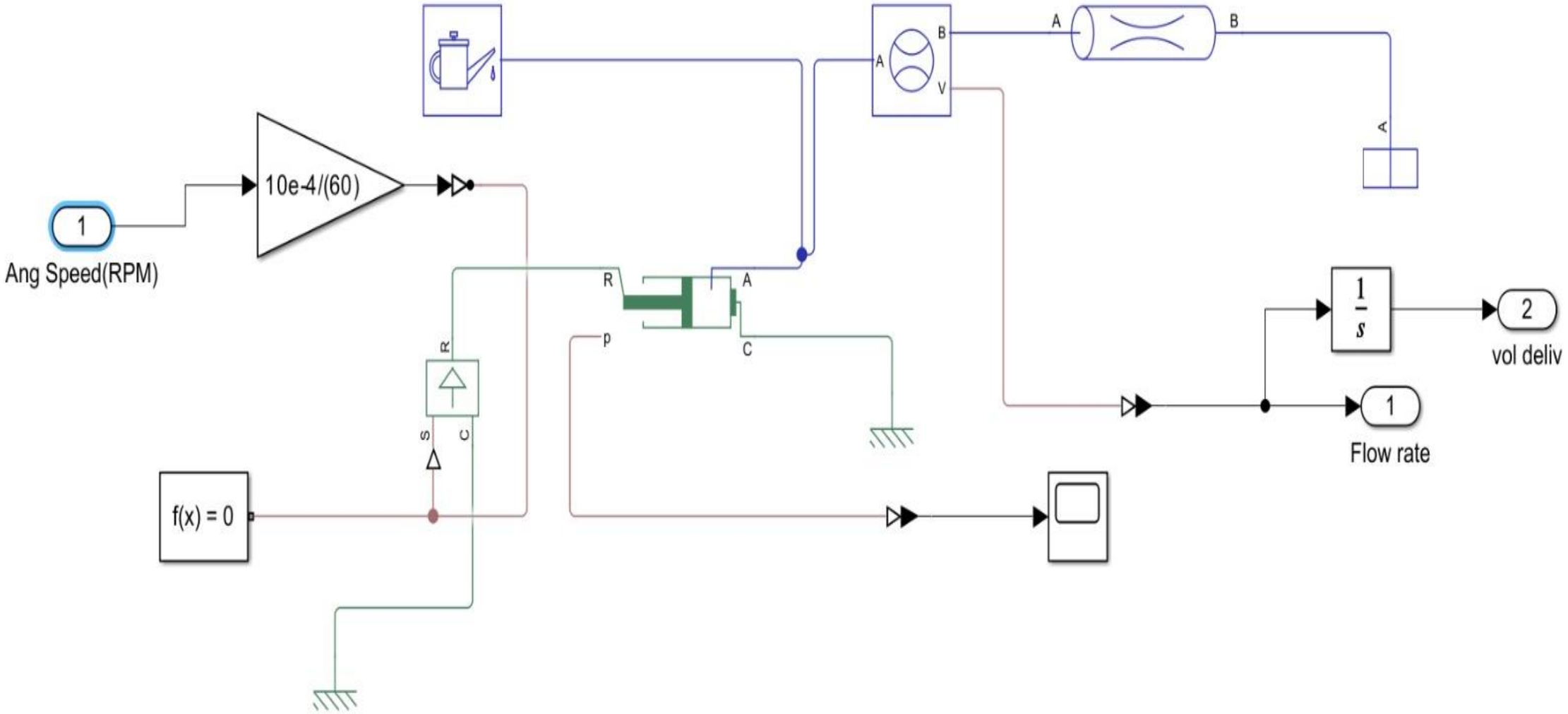
Top Model


WiFi
MOTOR HARDWARE
Position
Ref
Speed
Motor Hardware
MQTT SEND
ARDUINO
IsNew
Data
medical/pump/CIIP001/host
msg
target_speed
parse_mqtt
PUMP
Flow rate
Ang Speed(RPM)
vol deliv
Pump Dynamics
Vol Flow rate
total vol
speed
msg
publish_speed
ARDUINO
status
medical/pump/CIIP001/data

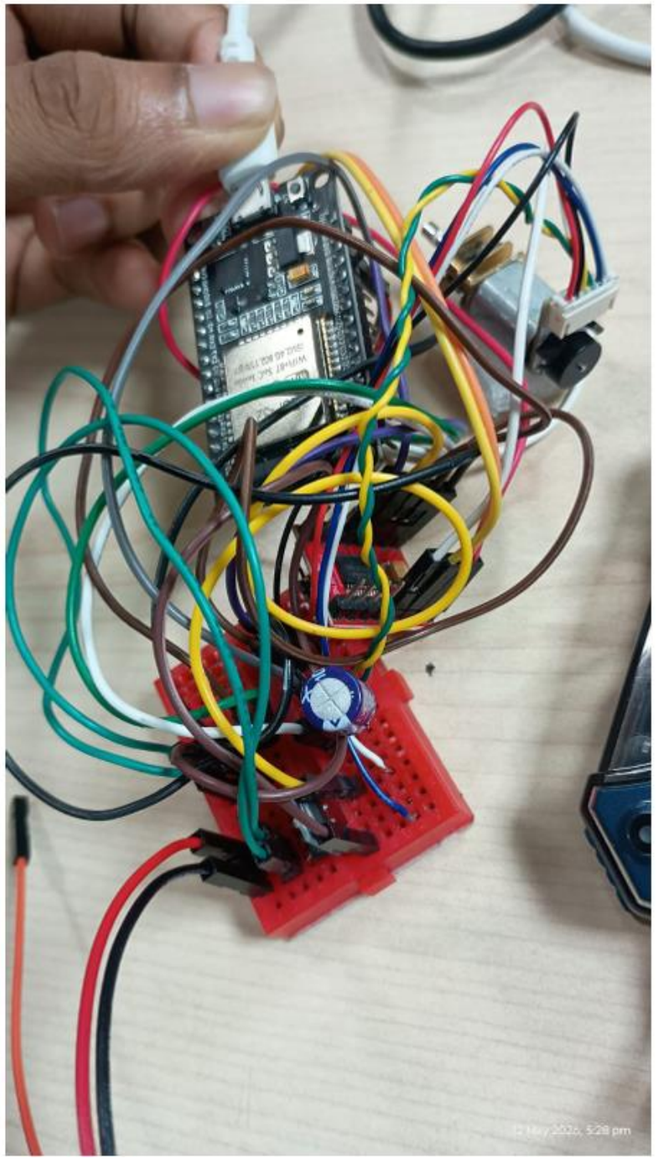
Motor Subsystem in Simulink

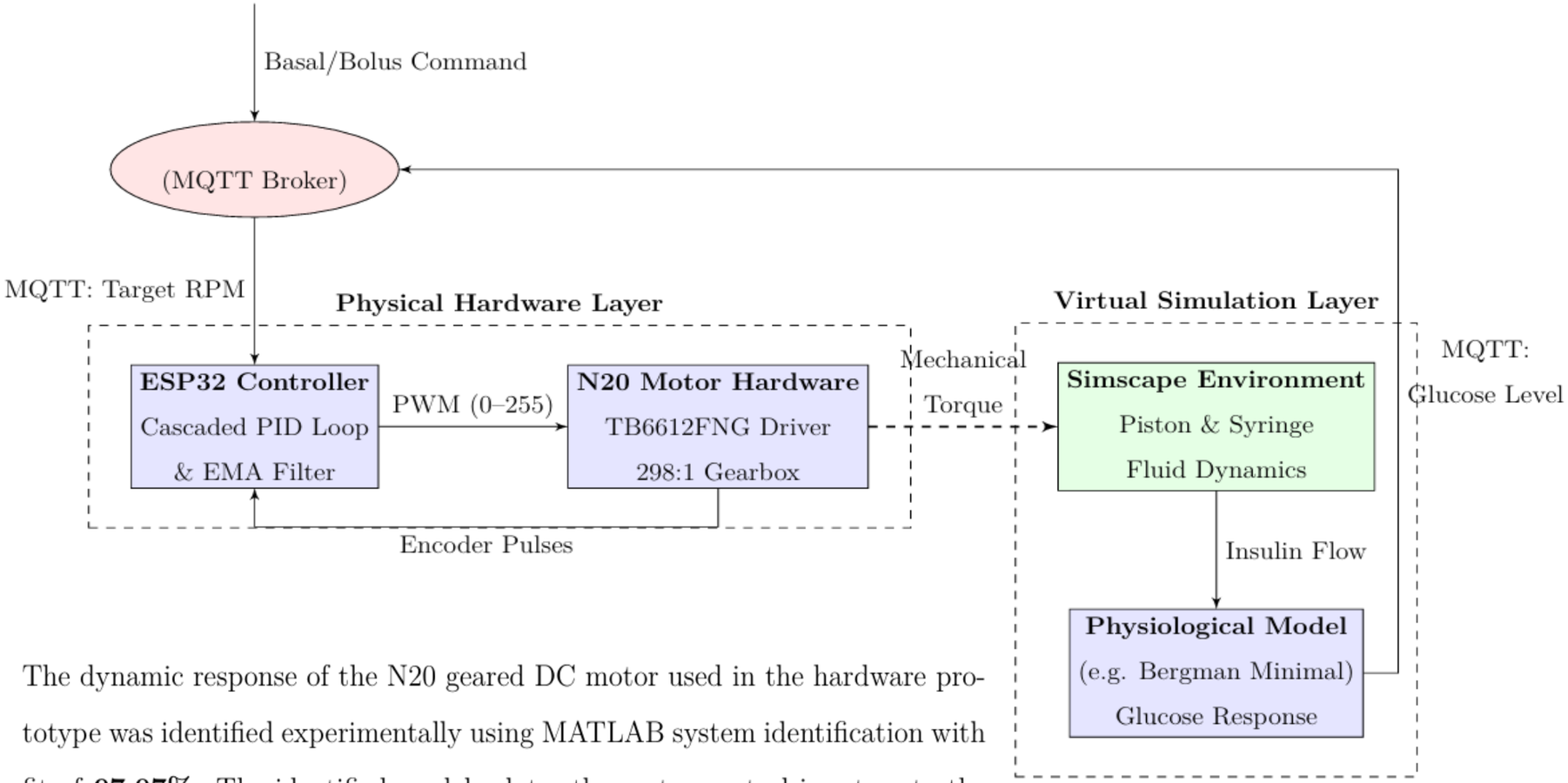


Pump dynamics model in Simulink



Hardware Model





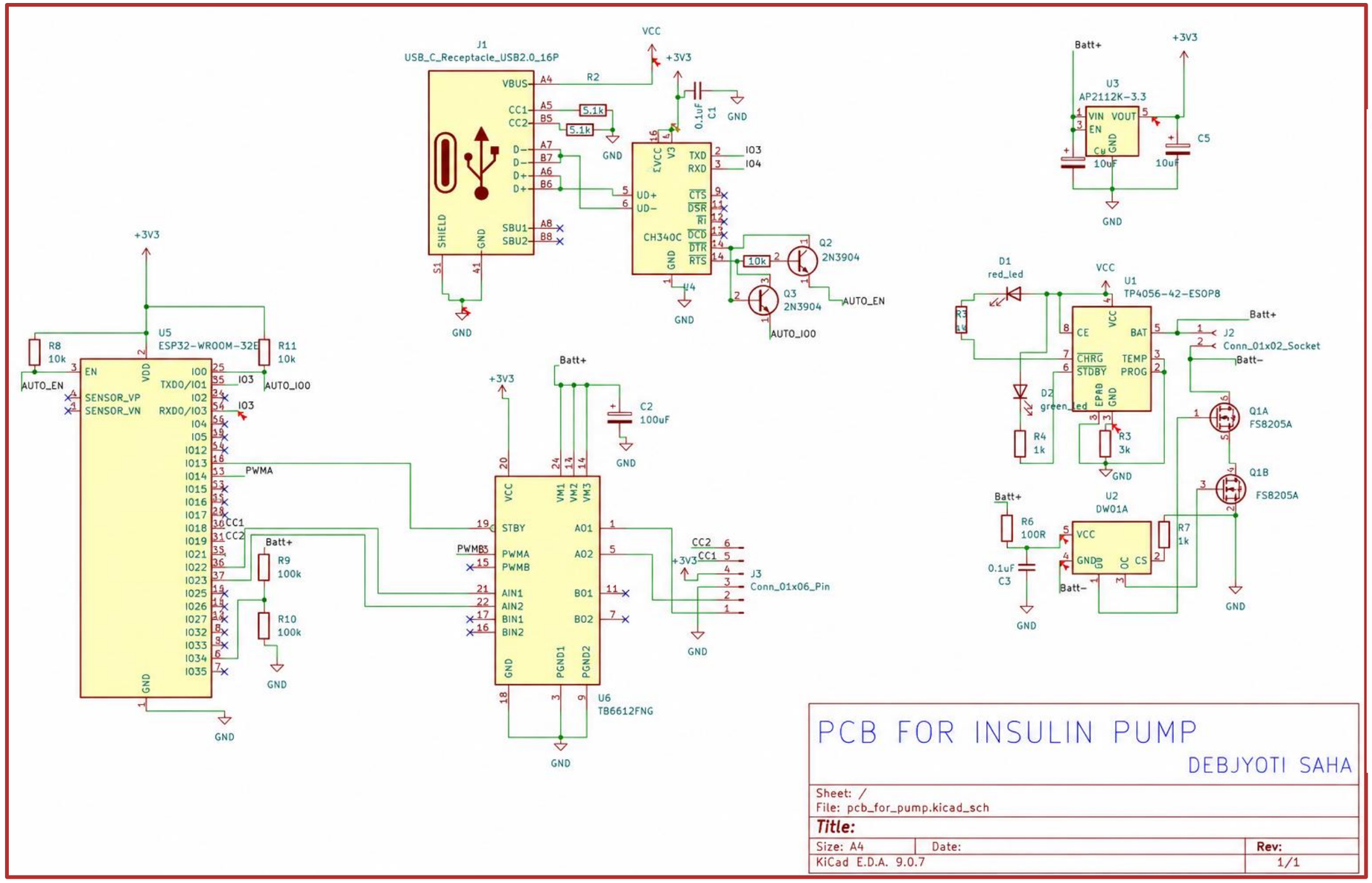
The dynamic response of the N20 geared DC motor used in the hardware prototype was identified experimentally using MATLAB system identification with fit of **97.07%**. The identified model relates the motor control input u_1 to the measured motor output y_1 and is represented as a discrete-time transfer function with a sampling time of $T_s = 0.01\,\text{s}$. The obtained transfer function is:

$$G(z) = \frac{Y_1(z)}{U_1(z)} = \frac{0.0006429}{1 - 1.801z^{-1} + 0.8063z^{-2}}.$$

Simulink Closed Loop Autotuner is used for

Obtaining K_p, K_i values with 99.97% convergence which are 28.43 and 108.1 respectively.

Schematic



PCB FOR INSULIN PUMP

DEBJYOTI SAHA

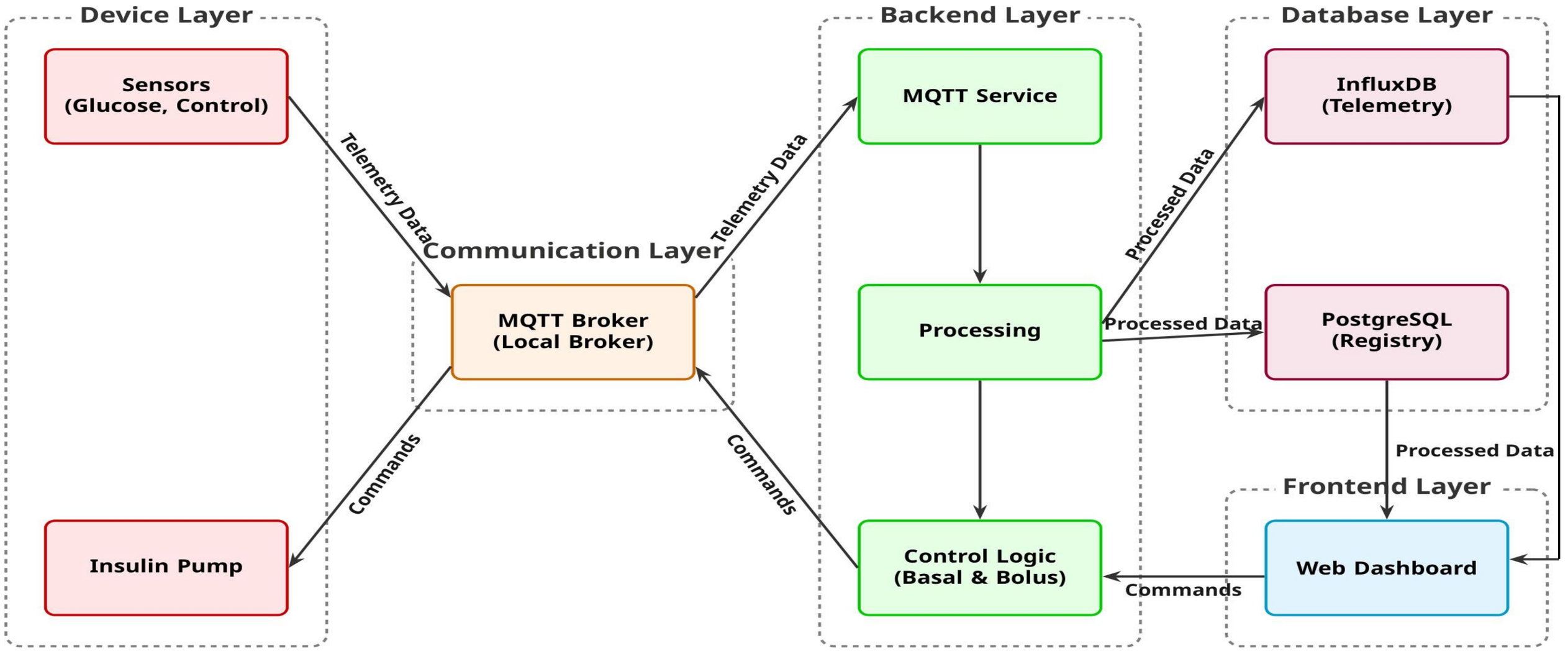
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Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		1/1

Sl. No.	Component	Qty.
1	ESP32 DevKit V1	1
2	TB6612FNG motor driver module	1
3	N20 metal gear motor with Hall encoder	1
4	Lead screw and syringe drive mechanism	1
5	5 V to 3.3 V voltage regulator	1
6	18650 Li-ion battery pack	2 cells
7	100 µF electrolytic capacitor	1
8	Syringe reservoir	1

WebApp Design



WebApp Design

Closed-Loop Insulin Infusion Pump Data Flow Overview :

1. Data Collection and Transmission (Device → Backend)

Continuous Monitoring

Sensors continuously track physiological parameters
Includes glucose levels and device status

Telemetry Data Transmission

Data is transmitted using the MQTT over local network
Direct communication between device and web app
Ensures low-latency and reliable message exchange

Backend Ingestion

MQTT Service subscribes to incoming data
Telemetry is received and forwarded for processing

2. Data Processing and Storage (Backend → Database)

Data Processing

Raw data is cleaned and formatted
Processing layer prepares data for storage and analysis

Time-Series Storage (InfluxDB)

Stores continuous sensor data over time
Optimized for high-frequency telemetry

Relational Storage (PostgreSQL)

Stores structured data such as:
- Device registry
- Patient information & System state logs

WebApp Design

Closed-Loop Insulin Infusion Pump Data Flow Overview :

3. Pump Control (Backend → Pump)

Command Generation and Dispatch

Control Logic generates machine-level commands
Commands are sent to the MQTT Broker for delivery

Insulin Delivery

Insulin Pump subscribes to its command topic
Receives instructions from MQTT
Adjusts insulin infusion rate accordingly

4. User Monitoring and Intervention

Real-Time Dashboard

Displays processed
Visualizes glucose levels, pump status, and history

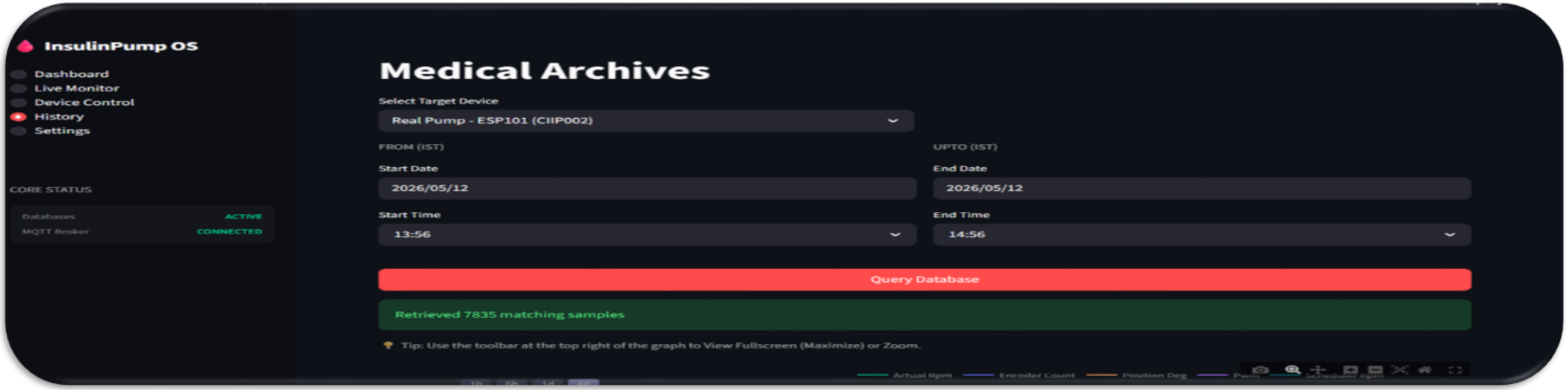
Manual Control

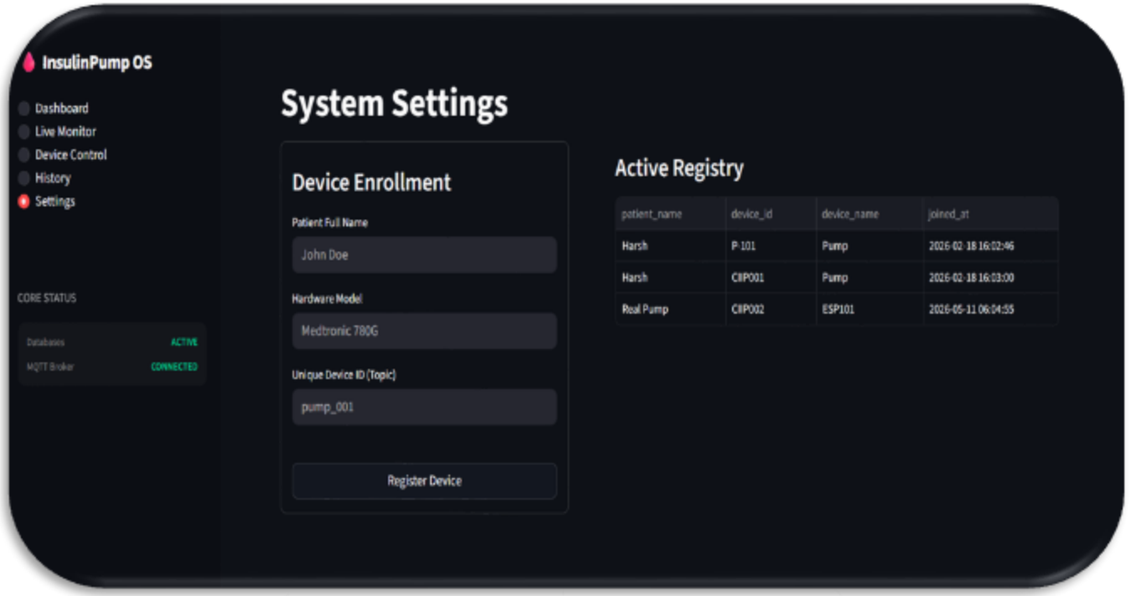
Users can input commands (e.g., bolus dose, settings adjustment)
Enables human intervention when required

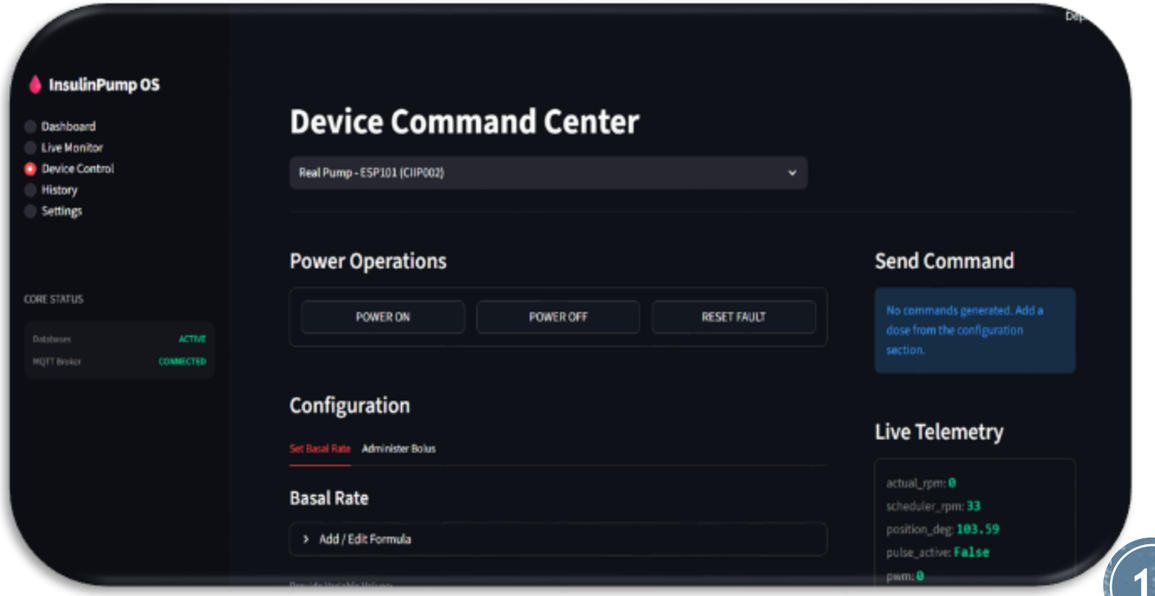
WebApp Design



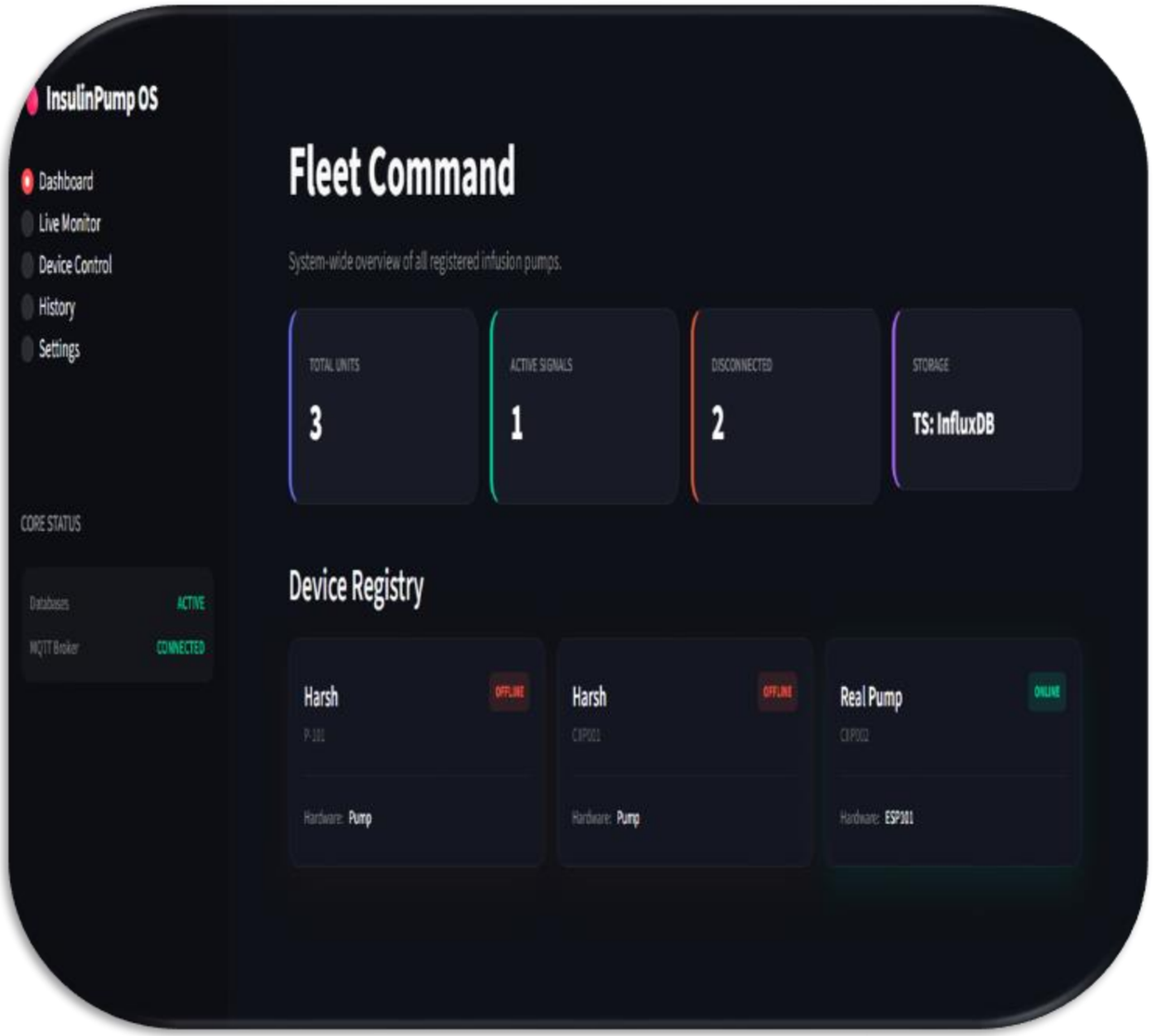
WebApp Design

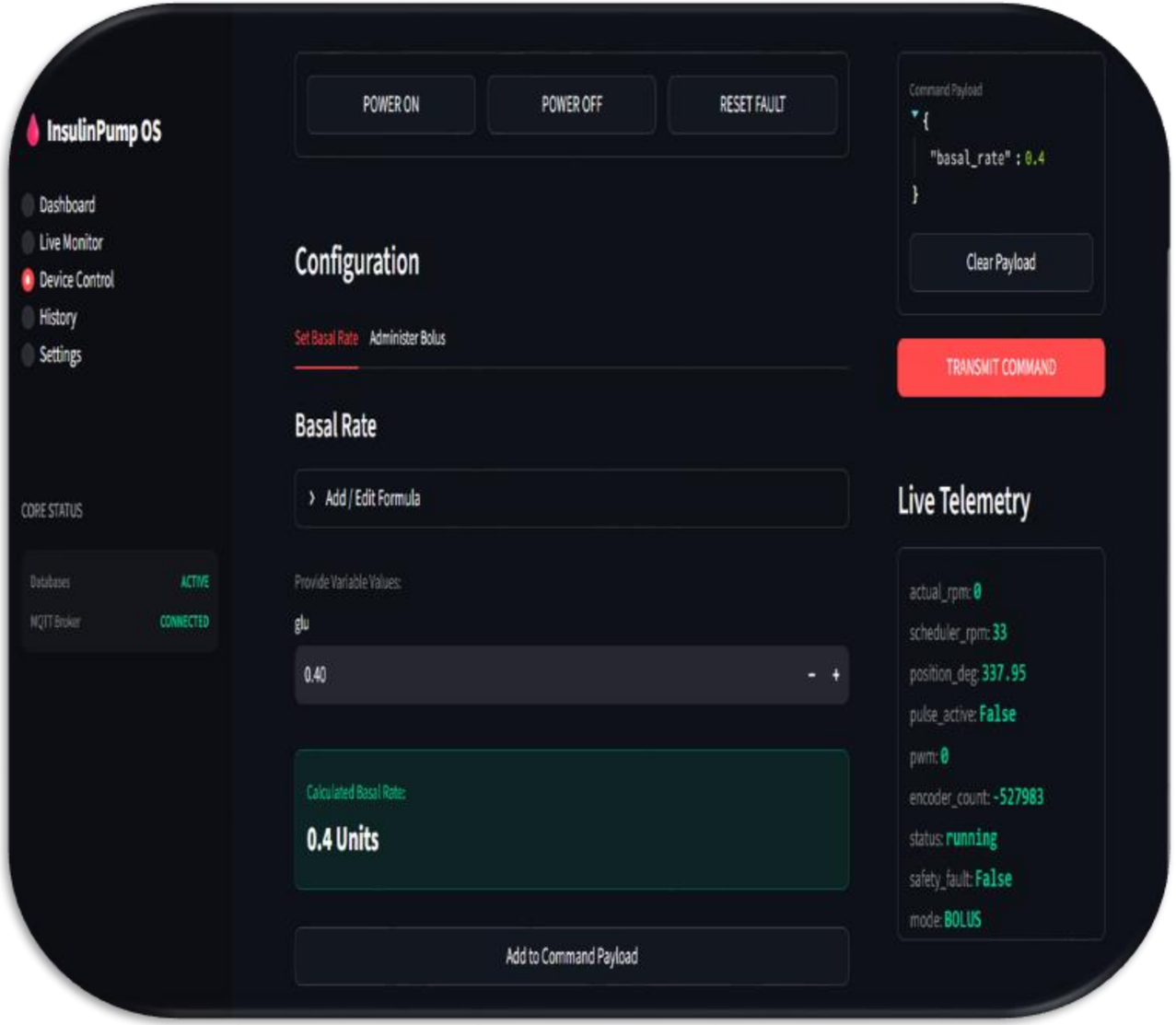




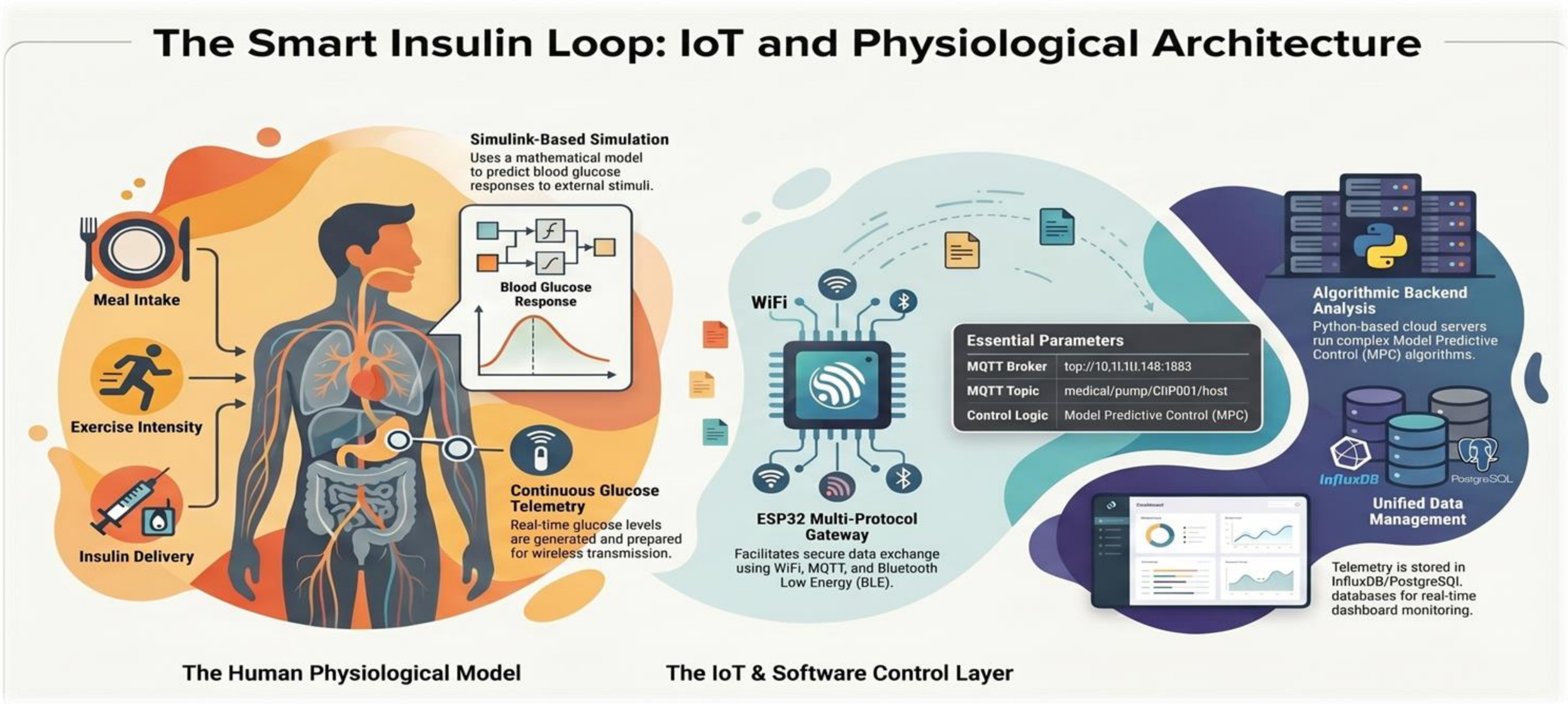


WebApp Design

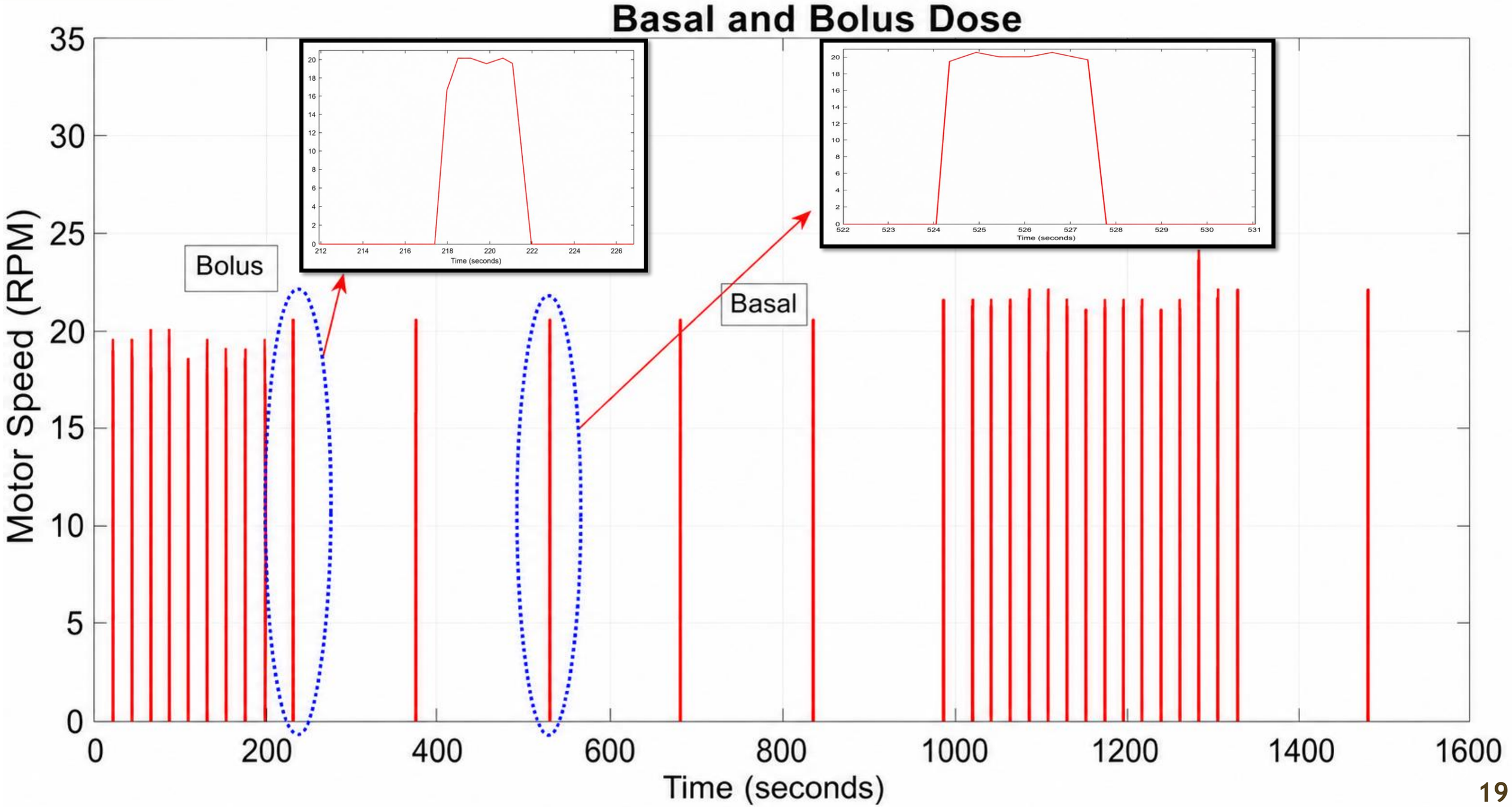




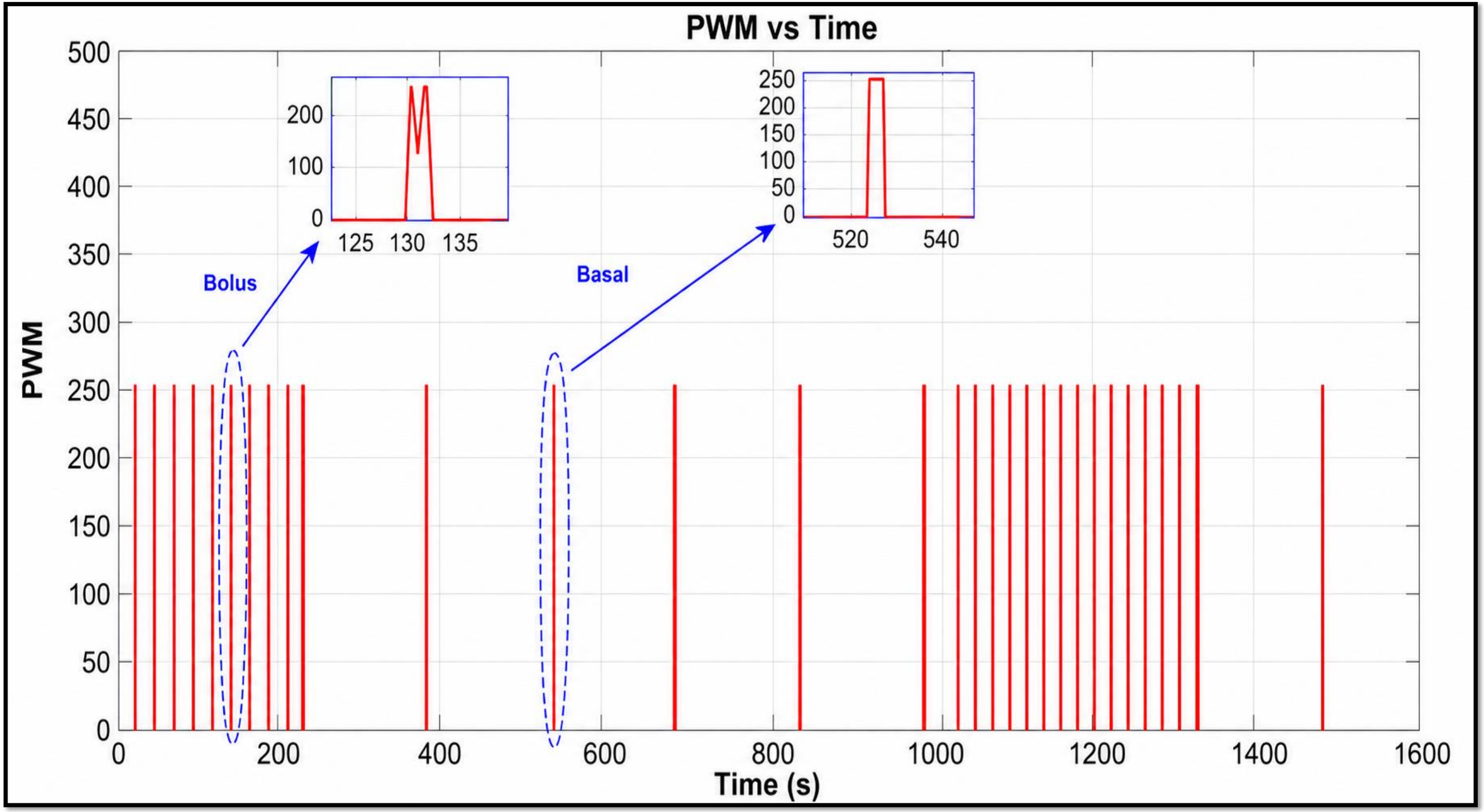
Complete Overview



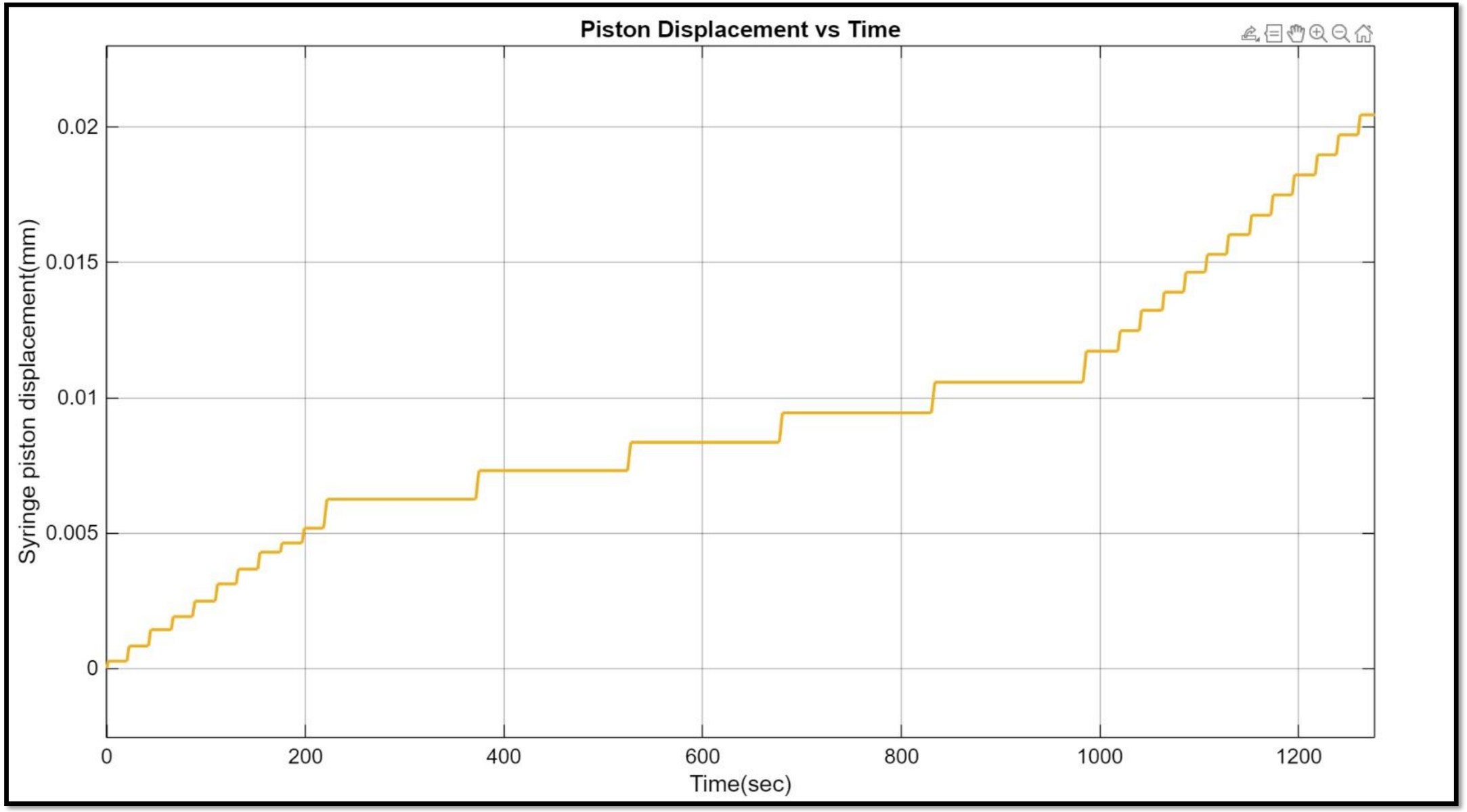
Results:


Basal and Bolus Dose
Bolus
Basal
Motor Speed (RPM)
Time (seconds)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
1400
1600
Time (seconds)
Time (seconds)

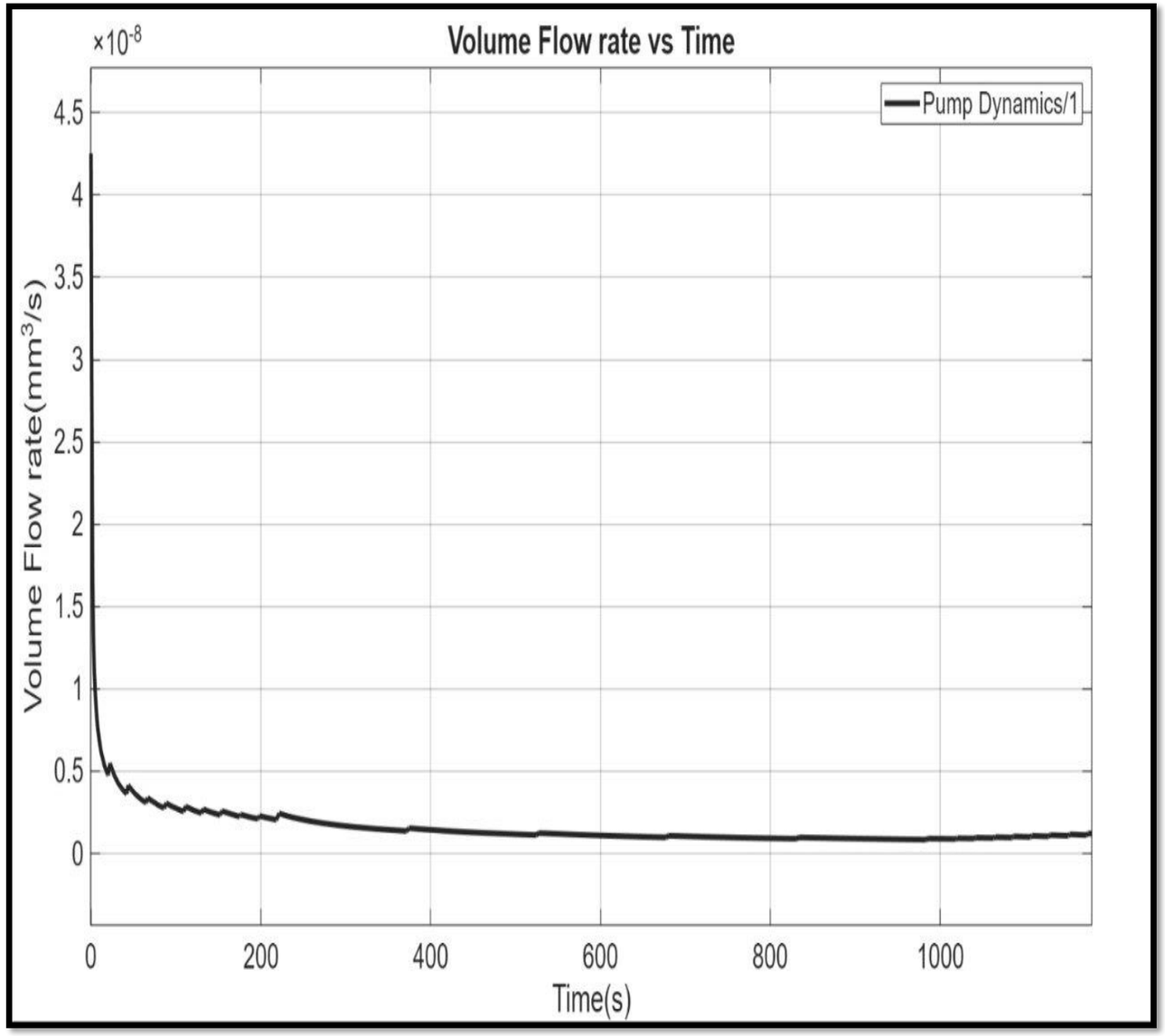
Results:

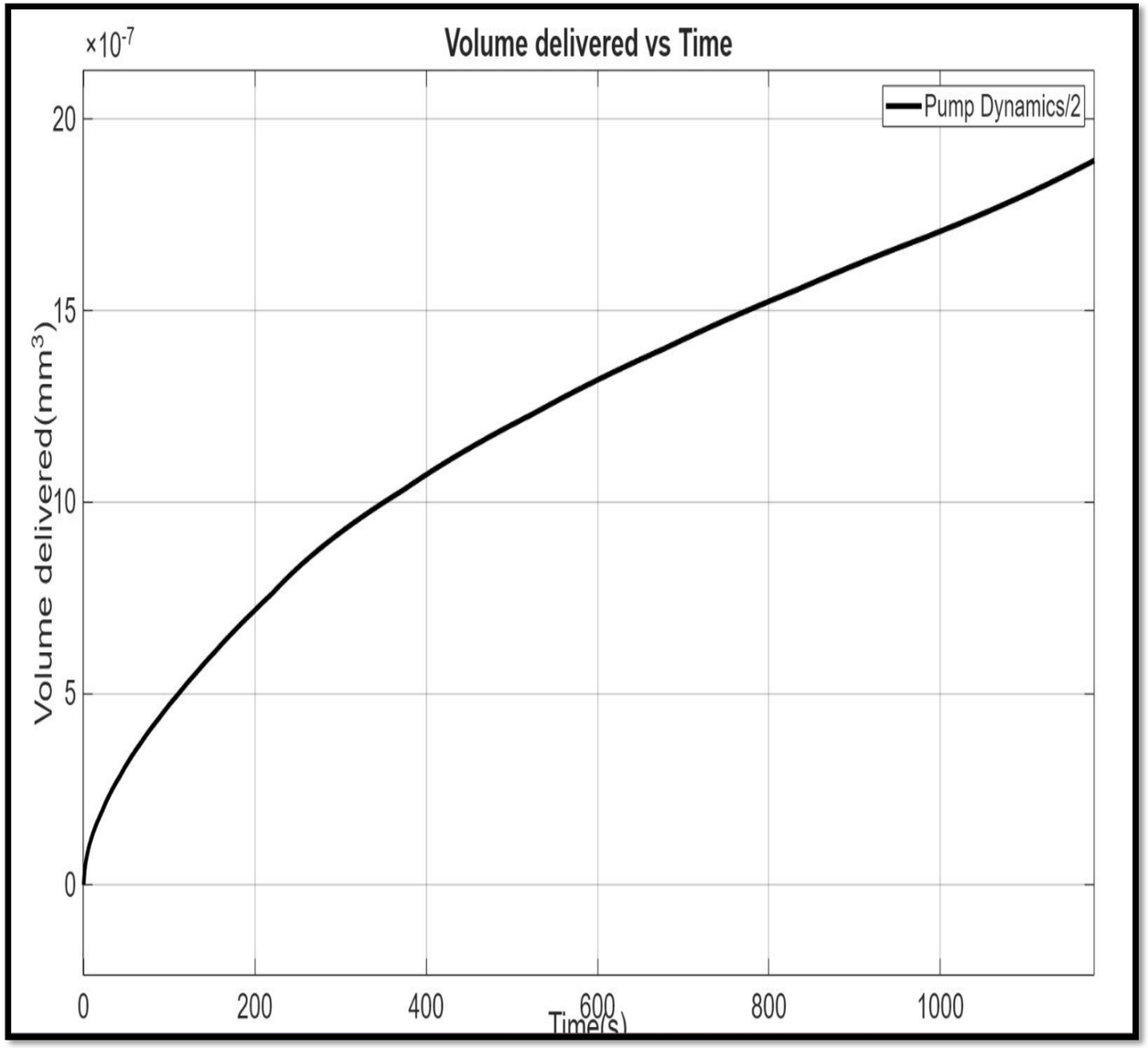

PWM vs Time
PWM
Time (s)
Bolus
Basal
0
50
100
150
200
250
300
350
400
450
500
0
200
400
600
800
1000
1200
1400
1600
125
130
135
520
540

Results:


Piston Displacement vs Time
Syringe piston displacement(mm)
0.02
0.015
0.01
0.005
0
0
200
400
600
800
1000
1200
Time(sec)

Results:


Volume Flow rate vs Time
×10-8
Pump Dynamics/1
Volume Flow rate(mm3/s)
Time(s)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
0
200
400
600
800
1000


Volume delivered vs Time
×10-7
Pump Dynamics/2
Volume delivered(mm3)
Time(s)
0
5
10
15
20
0
200
400
600
800
1000

Conclusion: Democratizing Insulin Pump Therapy

- Developed a **low-cost IoT-enabled** insulin infusion pump using ESP32, motor-driven syringe mechanism, and web-based monitoring.
- Integrated MQTT communication for remote monitoring, control, and scalability across multiple pumps.
- Prototype demonstrates the feasibility of an **affordable and intelligent insulin delivery platform**.

Future Development:

Closed-Loop Improvements

•Implement advanced controllers such as MPC, adaptive PID, and robust control for improved stability and accuracy.
•Develop adaptive insulin dosing algorithms based on patient response and glucose trends.
•Add adaptive gain scheduling to optimize controller performance.

Software & Embedded Enhancements

•Add automatic meal detection and predictive glucose control using CGM data.
•Perform long-term reliability testing and clinical validation for practical medical use.

System Expansion & Connectivity

•Design a custom PCB and compact portable enclosure.
•Develop custom PCB and final hardware casing with charging support.
•Integrate additional safety features like occlusion, leakage, and low-battery detection.

Thank You